



Innovative resources driven artificial intelligence orientation: The moderating role of environmental and executives' characteristics

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ABSTRACT

In the era of rapid artificial intelligence (AI) technology advancement, enterprises increasingly consider AI as a strategic imperative. However, the determinants of AI adoption and strategy formulation in enterprises remain elusive. Drawing on an integrated Resource-Based View (RBV) and Strategic Choice Theory (SCT) framework, this study presents a theoretical model that scrutinizes the factors and boundary conditions influencing enterprises in crafting AI strategies. Empirical assessments of theoretical hypotheses were conducted utilizing survey data gathered from 717 executives across global firms. We found strong evidence that internal innovation resources (IIR) and collaborative innovation networks (CIN) have a positive impact on firm's AI orientation. The study revealed that environmental uncertainty (EU) can negatively moderate the relationship between internal innovation resources, collaborative innovation networks, and AI orientation. A positive moderating effect was observed between corporate digital culture (DC) and the relationship between internal innovation resources, collaborative innovation networks, and AI orientation. Meanwhile, we found a positive moderating effect of Executives' Not-Invented-Here Syndrome (NIHS) on the relationship between internal innovation resources and AI orientation. This research contributes to the ongoing discourse on AI strategy in management and offers valuable guidance for organizations seeking to harness the potential of AI in a rapidly changing business environment.

1. Introduction

The integration of artificial intelligence (AI) technologies has become a crucial driver of management, planning, and operations. Approximately 80% of large companies have incorporated AI into their core business, marking a 70% increase over the past five years (Makarius et al., 2020). AI also represents a transformative trend shaping the future of emerging enterprises (Ding et al., 2014; Li et al., 2023). Xiaomi Group, for instance, has adopted a “software + hardware + artificial intelligence” strategy since 2016, demonstrating how enterprises can utilize AI and large data models to enhance consumer experiences and create sustained value across platforms, products, and services, surpassing traditional IT limitations (Gregory et al., 2021). The concept of artificial intelligence orientation reflects a strategic shift within enterprises, highlighting a commitment to leveraging AI for transformative purposes, including its adoption for intelligent transformation (Li et al.,

2021). By utilizing advanced algorithms for data analysis, organizations can uncover market trends, identify investment opportunities, and detect unique patterns in customer and transactional data (Mikalef et al., 2020). This AI-driven orientation empowers enterprises to extract actionable insights from vast datasets, facilitating informed decision-making and bolstering competitive advantage (Garbuio & Lin, 2019).

Artificial intelligence, hailed as a disruptive innovation that transcends human cognitive limitations (Balasubramanian et al., 2022; Cooper et al., 2023), necessitates a comprehensive exploration of the factors influencing AI adoption and the concurrent formulation of strategic initiatives (Chen et al., 2021). Regrettably, this crucial domain remains largely unexplored, resulting in enterprises lacking a robust framework for the formulation of AI strategies. Consequently, ambiguity persists regarding the effectiveness of enterprises in devising AI strategies (Makridakis, 2017). These circumstances underscore the urgent

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necessity for rigorous examination and in-depth analysis of the forces driving the orientation of artificial intelligence within enterprises. Such research can guide enterprises in seamlessly integrating AI into their strategic frameworks, enabling them to fully leverage AI to gain competitive advantages, enhance operational efficiency, and strengthen preparedness for future challenges.

This study aims to explore the drivers and boundary conditions that shape firms' AI orientation, utilizing an integrated framework based on the resource-based view (RBV) and strategic choice theory (SCT) as its primary analytical lens. RBV argues that an organization's strategic resources are essential in determining its strategic direction and creating sustained competitive advantages (Barney, 1991). Within the realm of artificial intelligence, firms possess valuable internal innovation resources, including proprietary datasets, internally developed algorithms, and specialized talent. Furthermore, they harness network resources to facilitate collaborative innovation across data, technology, and algorithm domains. These strategic resources form the foundation for sustainable growth and asset accumulation, as highlighted by Pan et al. (2018). Investigating the role of internal innovation resources and collaborative innovation networks is essential for understanding the drivers of AI orientation, offering valuable insights into the key factors shaping its development.

However, while the resource-based view provides valuable insights into how firm-specific resources shape AI orientation, it has notable theoretical limitations, as it predominantly focuses on internal resources and capabilities, often neglecting the broader strategic context. As Peng (2001) underscores, strategic decision-making is influenced by a broader array of external elements that RBV often overlooks. To bridge this research gap, our study integrates strategic choice theory into the theoretical framework, emphasizing the pivotal roles of top management teams and external environmental factors - including market dynamics and technological shifts - in determining strategic outcomes (Child, 1972). Strategic choice theory posits that strategic decision-making is inherently subjective and dynamic, shaped by both managerial characteristics and environmental factors. It highlights that strategic actions aim to mitigate environmental constraints, and are guided by the cognitive frameworks of decision-makers and the organizational culture (Child, 1997). Additionally, the dynamism of external forces means that effective organizational adaptation depends on the perceptual accuracy of decision-makers and their ability to respond appropriately to environmental conditions (Judge et al., 2015). This theory addresses a pivotal gap by elucidating how environmental factors and executive characteristics accommodate dynamic, unpredictable environments, thereby facilitating the successful implementation of AI strategies. This integration augments the depth and breadth of our analysis.

Hence, our study proposes that environmental factors and top-level management characteristics serve as critical boundary conditions influencing how an organization leverages internal innovation resources and external network resources in shaping its artificial intelligence orientation. Concerning environmental factors, the presence of environmental uncertainty emerges as a pivotal external variable with profound implications for enterprises. The turbulent nature of an uncertain environment amplifies risks and unpredictabilities, constraining the availability of resources and making resource allocation more intricate (Foss et al., 2013). Within the organization's internal environment, the role of digital culture assumes paramount importance. It enhances employees' awareness and sensitivity to digital technology, equipping the enterprise with the essential information and cultural foundations necessary for developing an AI strategy that thrives within an enabling environment (Nambisan et al., 2017). With regard to the attributes of top-level management, we argue that the prevalence of the "Not-Invented-Here Syndrome" (NIHS) among top managers - manifesting as a resistance toward integrating external ideas, solutions, or innovations within individuals or teams—has the potential to hinder opportunities for external and collaborative innovation. Consequently, this constraint

restricts the diversity, flexibility, and innovative capacity of organizational strategies, ultimately undermining the organization's capacity to effectively leverage innovative resources in pursuit of an AI orientation (Arias-Perez & Velez-Jaramillo, 2022a). Integrating the above diverse views, our research constructs a theoretical model that incorporates resource-based view and strategic choice theory to answer two key questions: (1) Do corporate innovation resources and networks drive the formulation of AI orientation? (2) How do environmental uncertainty, digital culture, and executives' NIHS affect the relationship between innovation resources and AI orientation? We conducted a global survey of 717 senior managers to address the above research questions.

This study enriches the theoretical landscape by constructing a comprehensive model to examine the drivers and boundary conditions of artificial intelligence orientation. Through an in-depth analysis of the AI orientation concept and the development of a measurement scale, the study enhances both the theoretical foundation and strategic dimensions of AI. By integrating the resource-based view and strategic choice theory into a unified framework, it addresses a critical research gap in understanding the determinants and constraints of AI orientation, thereby increasing the framework's relevance for innovation strategy in the digital era. From a managerial standpoint, these contributions possess immediate relevance. Firms increasingly acknowledge AI as a strategic imperative, prompting substantial investments in AI technologies to enhance core competitive advantages. Organizations with robust internal innovation resources and collaborative networks are better poised for effective AI strategy formulation and implementation. Moreover, the successful development of an AI strategy necessitates fostering forward-thinking, open-minded decision-makers to ensure alignment with the dynamic internal and external environments. These managerial implications offer valuable guidance for enterprises aiming to capitalize on the opportunities presented by AI in today's rapidly evolving business landscape.

2. Theoretical framework and hypothesis development

2.1. Artificial intelligence as a strategic direction

As AI takes on an increasingly vital role within organizations, scholars have recognized its importance in enhancing operational capabilities and sustaining competitive advantage. They emphasize the need for firms to strengthen their AI capability—the ability to select, orchestrate, and leverage AI-specific resources (Chen et al., 2022; Mikalef & Gupta, 2021). Firms must also achieve AI assimilation, which refers to the integration of AI across organizational dimensions, embedding it into project activities and work processes until it becomes routine (Wamba, 2022). Notably, AI has become a strategic imperative for businesses, profoundly influencing operations. Companies are now aware of the necessity for a clear AI strategy to manage the complex applications of this transformative technology (Helfat et al., 2023). AI orientation serves as the foundation of a strategic framework, covering the assimilation of AI technology and strategic decision-making regarding investments, adoption, and AI-focused management practices (Li et al., 2021). This alignment enhances AI investment decisions, facilitates AI's application to business challenges, and offers stakeholders a transparent rationale for AI deployment. Consequently, AI orientation maximizes AI's potential while carefully managing its associated risks (Fountain et al., 2019).

AI orientation has evolved, with organizations now positioning it as the linchpin of their digital transformation initiatives. From machine learning paradigms to robotic process automations, AI components have become integral to the IT infrastructure (Brynjolfsson et al., 2019). This symbiosis augments enterprises' capabilities in data analytics, fortifies network security, and enhances cloud computing proficiencies. AI-driven analytics refine server utility, data stewardship, and application efficacy (Davenport & Ronanki, 2018). In contrast to traditional IT paradigms centered on data acquisition and processing, AI encapsulates

auto-regenerative learning, enabling autonomous decision-making on behalf of stakeholders (Faraj et al., 2018). However, due to its relative newness, AI lacks mature assessment benchmarks, compelling managers to devise metrics for evaluating AI's decision-making capabilities and its effectiveness in managing organizational processes (Rudin, 2019). The absence of these metrics introduces uncertainties into assessing AI's reliability (Makridakis, 2017).

Simultaneously, AI strategy embodies an innovation strategy with the potential to reshape products, services, and operational procedures. For instance, AI-powered innovations are propelling autonomous vehicles in the automotive industry and enhancing patient care in healthcare by predicting ailments and aiding in diagnoses (Garbuio & Lin, 2019; Haenlein & Kaplan, 2019). AI can automate mundane tasks and forecast operational challenges, leading to innovative operational blueprints that boost productivity (Bughin & Van Zeebroeck, 2018). AI-led strategies can also foster disruptive business models, exemplified by platforms like Uber and Airbnb, which utilize AI for dynamic pricing and optimization (Von Krogh et al., 2023). Leveraging AI for innovation equips businesses to respond agilely in a dynamically competitive environment and manage the deluge of data efficiently (Uren & Edwards, 2023). Additionally, infusing the innovation trajectory with AI can enhance enterprise value while mitigating the risks and financial implications of the innovation journey (Haefner et al., 2021).

Despite the growing recognition of the importance of AI in the business landscape, there remains a dearth of comprehensive research that delves into the factors that guide and shape the development of AI strategies (Chen et al., 2021). This knowledge gap presents a significant challenge for organizations seeking to effectively harness the potential of artificial intelligence. Addressing this research void and exploring the driving factors behind enterprises' formulation of artificial intelligence strategies is of paramount importance. Such exploration provides a roadmap for organizations to navigate the complexities of AI adoption, investment, and integration into their existing processes. Understanding these factors empowers decision-makers to make informed choices, allocate resources judiciously, and align AI strategies with overarching business goals. It enables organizations to optimize the benefits of AI technology while minimizing risks and uncertainties.

2.2. Theoretical framework integrating RBV and SCT

The resource-based view (RBV) offers a robust framework for comprehending the pivotal role of internal innovation resources and collaborative innovation networks in shaping Artificial Intelligence (AI) orientation. RBV asserts that core internal attributes, encompassing management competencies and technological proficiency, serve as fundamental drivers of sustained development (Helfat et al., 2023; Wu et al., 2008). At the core of RBV lies the concept of strategic resources, essential for fostering continuous growth through the assimilation of both internal and external innovation assets (Pan et al., 2018). These strategic resources possess attributes such as tacitness, complexity, exclusivity, and gradual accumulation. Moreover, they must also fulfill the criteria of being valuable, rare, and inimitable, thereby resisting easy replication by competitors (Barney, 1991).

In the ever-evolving business landscape, AI has become integral to operations, products, and services. The innovative use of AI capabilities renders them valuable, rare, and difficult for competitors to imitate, highlighting the strategic importance of AI orientation in securing and sustaining competitive advantage (Teece, 2018). While many firms may possess similar AI tools, it is their distinctive integration and application that truly differentiates them. AI strategy extends beyond technology, encompassing skilled human capital and collaborative networks that foster resource synergy and innovation (Speckbacher et al., 2015). These AI-related skills are intangible strategic resources, offering a sustainable competitive edge that rivals find hard to replicate (Chowdhury et al., 2023). This approach aligns seamlessly with RBV, which emphasizes the role of both internal and network resources in sustaining competitive

advantage and shaping effective AI strategies (Cai et al., 2016).

While the RBV provides insights into AI strategy development, it has notable limitations in dynamic environments. Its emphasis on firm-specific, inimitable resources (Barney, 1991) overlooks the fluid nature of AI-driven competition. By treating resources as static, RBV underestimates the need for continuous reconfiguration amid technological shifts and evolving market conditions. Given AI's rapid advancements, firms must not only acquire AI resources but also develop dynamic capabilities to integrate and reconfigure them effectively (Teece, 2018). Moreover, it fails to account for managerial cognition and strategic agency, which are crucial for AI adoption and adaptation (Helfat & Martin, 2015). RBV also neglects external factors such as regulatory shifts and ecosystem collaborations, making it incomplete for AI strategy formulation in today's volatile business landscape (Peng, 2001).

Strategic choice theory (SCT) complements this gap by emphasizing the joint influence of leadership agency and external environmental dynamics in shaping strategic decisions (Child, 1972). SCT emphasizes the proactive and iterative nature of strategic decision-making, wherein managerial perceptions and responsive actions play a central role (Judge et al., 2015). At its core, SCT underscores that strategic decisions are not solely dictated by external constraints but are shaped by executive cognition, internal philosophies, and strategic intent (Montanari, 1979). Organizations do not merely adapt to their environment; rather, leaders actively interpret, frame, and recalibrate strategic responses based on cognitive biases, embedded ideologies, and cultural norms (Child, 1997). These cognitive frameworks function as pre-established lenses through which decision-makers perceive opportunities and threats, influencing the direction of strategic choices. Furthermore, SCT highlights the dynamic and reciprocal relationship between organizations and their external environments. Environmental forces impose structural constraints, yet strategic leaders retain significant subjective agency in navigating and reshaping these conditions (Judge et al., 2015). This perspective reinforces the notion that organizational adaptation and strategic direction are not predetermined but rather emerge through a continuous negotiation between environmental contingencies and leadership-driven decision-making. Ultimately, the effectiveness of an organization's strategic response hinges on how well its decision-makers interpret environmental conditions, anticipate shifts, and execute adaptive strategies (Child, 1997).

Integrating the RBV with SCT provides a robust framework for analyzing AI strategy drivers and boundary conditions. RBV highlights the strategic importance of firm-specific resources, such as proprietary algorithms and machine learning expertise, which are valuable, rare, and difficult to imitate (Barney, 1991). However, in the dynamic AI landscape, resource possession alone is insufficient; firms must continuously adapt resource deployment to external market forces and technological shifts, a core tenet of SCT (Child, 1972). This underscores the need for real-time strategic recalibration to sustain competitive advantage (Fountain et al., 2019; Helfat et al., 2023). Additionally, while RBV emphasizes internal resources, it fails to fully account for the role of organizational culture in shaping how AI resources are leveraged. A firm's cultural orientation—openness to innovation, and risk tolerance—determines whether AI capabilities translate into strategic advantage or remain underutilized. SCT addresses this gap by emphasizing that firms with adaptive, innovation-driven cultures are more effective in transforming AI resources into business value (Judge et al., 2015). Finally, managerial mindset plays a decisive role in orchestrating AI strategies. While RBV focuses on resource possession, it does not sufficiently consider how executive cognition, strategic vision, and decision-making under uncertainty influence resource utilization. SCT fills this gap by highlighting how leaders' cognitive flexibility, risk perception, and strategic foresight determine AI adoption, deployment, and adaptation (Helfat & Martin, 2015). Executives with a proactive mindset are more likely to recognize AI's disruptive potential and dynamically align resources with evolving market conditions (Teece,

2018). Conversely, rigid or risk-averse leadership can stifle AI-driven transformation, leading to competitive inertia. Thus, the integration of RBV and SCT advances a more nuanced and dynamic understanding of AI strategy formulation.

This study tries to explore the driving factors and boundary conditions of AI orientation by construct a theoretical framework integrating RBV and SCT. This approach underscores the dynamic interplay between organizational resources and strategic decision-making, highlighting how environmental factors and managerial attitudes shape strategic choices in the increasingly complex landscape of AI deployment. From the RBV perspective, Internal innovation resources within enterprises play a pivotal role in catalyzing the formulation of AI strategies, contributing to the development of an AI orientation that effectively leverages these resources. Concurrently, the scope of RBV extends to include network resources, positing that resources originating from digital networks align harmoniously with RBV principles (Jiang et al., 2018). These network resources, shaped by path-dependent processes and societal intricacies, are acknowledged as relatively scarce and critical drivers of an enterprise's competitive advantage, thereby enhancing the organization's AI orientation (Jiang et al., 2015).

From the SCT perspective, our analytical framework is designed to explore how both the external environments and internal culture—such as environmental uncertainty, digital culture, and executives' mindset, including phenomena like the not-invented-here syndrome—affect the nexus between innovation resources and the formulation of AI strategies. Environmental uncertainty underscores the need for enterprises to proactively navigate unpredictable external factors, especially in the rapidly evolving AI landscape (Arias-Perez & Velez-Jaramillo, 2022a; Foss et al., 2013). Additionally, a strong digital culture enhances decision-making by fostering innovation and digital proficiency (Nambisan et al., 2017), equipping firms with the necessary knowledge for effective AI adoption. However, NIHS among executives can hinder adaptability by resisting external innovations, limiting collaboration, and reducing the flexibility of AI strategies (Antons & Piller, 2015; Foss & Foss, 2005; Hannen et al., 2019). Therefore, this integrated framework can help understand how organizations navigate the intricacies of AI strategy within a rapidly evolving landscape, ultimately can enhance their adaptability and innovation potential in the era of AI.

2.3. Innovative resources and artificial intelligence orientation

Innovative resources form the foundation for enterprises to build their innovation initiatives, encompassing diverse elements that drive creative outcomes and support business evolution (Pan et al., 2018). In the context of the knowledge economy, internal innovation resources have emerged as critical strategic assets for enterprises. The synergistic integration of these resources can significantly enhance an organization's innovation capabilities and efficiency, ultimately strengthening its innovative capacity (Zhang et al., 2018). Furthermore, it is widely acknowledged that collaborative innovation networks hold immense strategic value for enterprises (Lavie, 2006; Najafi-Tavani et al., 2018). These networks enable collaboration with external partners, facilitating the exchange of knowledge, ideas, and resources. Through the utilization of these networks, organizations can augment their overall innovation capability (Kim et al., 2021).

Internal innovation resources encompass innovative human resources, innovative financial resources, as well as technological and informational resources (Pan et al., 2018). Innovative human resources represent the R&D personnel engaged in technological innovation pursuits within enterprises. These individuals constitute the forefront of technological innovation activities and possess robust innovation capabilities (Collins & Smith, 2006). Innovation financial resources denote the capacity and scale of enterprises' investment in technological innovation initiatives, typically materializing as research and development funds. These funds provide the tangible foundation necessary for the advancement of technological innovation activities (Julian &

Ofori-Dankwa, 2013). Primary sources of innovative financial resources include government funding, corporate allocations, bank loans, and venture capital. Internal technological and informational resources encompass a myriad of knowledge and information resources that are internally accumulated to ensure the effective completion of enterprises' technological innovation endeavors (Zhang et al., 2018). This domain includes technical achievements, technological intelligence, competitive insights, and other relevant aspects.

The internal innovation resources within enterprises possess the capacity to effectively drive the formulation of strategies related to artificial intelligence. This assertion is supported by the following reasons: Firstly, the presence of highly skilled innovative human resources, such as Chief Information Officers (CIOs) and technology innovation teams, equips enterprises with a deep understanding of the untapped potential and application domains of artificial intelligence technology. This, in turn, facilitates a more nuanced evaluation of the opportune instances and methods for integrating AI (Li et al., 2021; Mikalef & Gupta, 2021). Simultaneously, robust innovation capabilities strengthen enterprises' proficiency in artificial intelligence, enabling exploration of new solutions and application scenarios within the artificial intelligence domain. The creativity exhibited by these innovative human resources plays a pivotal role in the realm of artificial intelligence strategy (Mikalef & Gupta, 2021).

Secondly, the availability of abundant innovative financial resources plays a dual role: it not only strengthens research and development efforts in the field of artificial intelligence, including scientific exploration, technological advancement, and product innovation but also facilitates the formulation and execution of artificial intelligence strategies (Van Zeebroeck et al., 2021). This approach simultaneously helps mitigate the risks associated with research and development and the deployment of artificial intelligence technology within enterprises. It allows for the exploration of emerging technologies and business models, thus laying the foundation for the core competitive advantages of enterprises (Dubey et al., 2020; Krakowski et al., 2023).

Finally, internal technological and informational resources are pivotal in amassing knowledge related to artificial intelligence technology, encompassing algorithms, data analytics methodologies, software development tools, and beyond. This knowledge accumulation equips organizations with the capability to forecast market trends and technological advancements, while also facilitating internal collaboration to accelerate the growth, dissemination, and adoption of artificial intelligence technology (Fountain et al., 2019).

In light of the above analysis, the present study postulates the following hypothesis:

H1. Internal innovation resources will have a positive effect on AI orientation.

As a powerful mechanism for driving strategic transformation, enterprises establish strategic networks with key stakeholders to collectively pursue common strategic objectives (Möller & Halinen, 2017). A collaborative innovation network represents an enterprise's engagement with a diverse range of partners, including suppliers, customers, competitors, and research organizations, with the goal of integrating complementary knowledge and competencies (Dittrich & Duysters, 2007). The scope of a collaborative innovation network includes activities such as enhancing distribution channels, diversifying product portfolios, increasing production adaptability, and complying with regulatory standards, among other aspects (Najafi-Tavani et al., 2018).

Collaborative innovation networks play a crucial role in shaping the formulation of artificial intelligence strategies within enterprises. Primarily, the collaborative innovation process, which involves a diverse range of stakeholders, serves as a key avenue for acquiring complementary knowledge and synergistic resources to drive innovation (Zhou & Li, 2012). In the context of artificial intelligence, different partners possess varying technological expertise, skills, and reservoirs of data. By establishing partnerships with suppliers, customers, competitors, and

research organizations, enterprises can access essential strategic resources, including technology, data, and human capital, thereby strengthening the foundation for the development and implementation of AI strategies (Nieto & Santamaría, 2007).

Secondly, collaborative innovation networks provide fertile ground for discovering new opportunities within the artificial intelligence landscape. Through partnerships with various collaborators, enterprises acquire knowledge and strategic resources from diverse domains, enhancing their understanding of the potential and limitations of AI technology (Zeng et al., 2023). By leveraging the contributions of partners and innovation networks, enterprises tap into scientific and technological advancements in the AI field to incubate innovative products, services, or business models. This, in turn, enables more precise and viable strategic decision-making (Ireland & Webb, 2007).

Thirdly, collaborative innovation networks serve to mitigate risks associated with R&D endeavors and promote cost-sharing imperatives. In the realm of artificial intelligence, both technology research and development, as well as its application, frequently necessitate substantial investments amid inherent uncertainties. Scholarly discourse underscores the notion that joint R&D ventures with partners effectively distribute risks and alleviate the exclusive burden borne by individual enterprises (Faems et al., 2005; Najafi-Tavani et al., 2018). This, in turn, establishes a resource-enriched foundation for the implementation of an AI strategy.

Based on the analysis presented thus far, this study posits the following hypothesis:

H2. *Collaborative innovation networks will have a positive effect on AI orientation.*

2.4. Environmental uncertainty as a moderator

Environmental uncertainty arises due to the inherent bounded rationality of managers, limiting their ability to comprehensively gather, process, and interpret information about their organizational environment. These limitations primarily result from the volatility and ambiguity present in the environment (Vecchiato, 2012). Essentially, when business decision-makers struggle to predict changes in the environment—encompassing factors like competitor dynamics, sociocultural trends, and technological shifts—the environment is considered uncertain (Arias-Perez & Velez-Jaramillo, 2022b; Haarhaus & Liening, 2020). Duncan (1972) identified complexity, stemming from the proliferation and diversity of external factors, and the speed of external change as key contributors to this uncertainty. Exponential technological advancement further exacerbates this situation, leading to frequent disruptive shifts, shorter product life cycles, and flexible business models.

Given the unpredictable nature of the modern business climate, it is crucial to develop strategies for navigating increasing environmental uncertainty (Haarhaus & Liening, 2020). However, traditional managerial practices often seem ill-equipped to deal with these rapid and unpredictable shifts (Haarhaus & Liening, 2020). From a resource-based perspective, an enterprise's core resources—both tangible and intangible—along with their operational capabilities, play a central role (Schilke et al., 2018). Robust perceptual abilities and early warning systems are essential for navigating volatile environments, ensuring the timely identification of emerging market opportunities and looming threats (Li & Liu, 2014).

This investigation posits that environmental uncertainty's inherent volatility can jeopardize strategic formulation. Specifically, in higher environmental uncertainty, the relationship between internal innovation resources and AI orientation maybe weaken on several reasons. First, as organizations encounter heightened unpredictability, they often adopt a risk-averse posture, prioritizing short-term stability over long-term investment in innovation (Roper & Tapinos, 2016). This conservative mindset can stifle the strategic commitment necessary for fostering AI initiatives, which rely heavily on substantial internal

resources such as R&D capabilities (Foss et al., 2013). Second, environmental dynamism can obscure the clarity needed for organizations to identify optimal investment opportunities in innovation. Frequent changes in market conditions create ambiguity, leading firms to divert resources towards immediate operational challenges rather than strategic AI pursuits (Ameje et al., 2023). Additionally, the complexities associated with AI—an evolving and multifaceted technology—can further complicate internal decision-making processes. Under conditions of uncertainty, firms may feel pressured to respond quickly, which can erode confidence in their ability to predict the outcomes of AI integration (Daft et al., 1988; Haarhaus & Liening, 2020). Consequently, the cumulative effect of these dynamics suggests that as environmental uncertainty escalates, the effectiveness of internal innovation resources in promoting AI orientation diminishes, necessitating a more nuanced understanding of resource allocation strategies.

In addition, collaborative innovation networks play a vital role in harnessing complementary capabilities between firms, enhancing the development of innovative AI solutions. However, as environmental uncertainty rises, organizations may shift their focus towards immediate resilience, reducing their commitment to collaborative efforts (Ameje et al., 2023). This shift weakens the collective drive needed to pursue joint AI initiatives, limiting the advantages of collaboration. Additionally, the dynamic nature of the environment complicates communication and alignment among network partners, making it difficult to coordinate efforts towards shared objectives. Frequent changes may disrupt established roles, leading to inefficiencies in resource sharing and decision-making (Issa et al., 2022). The complexity of AI technologies further exacerbates these challenges, as partners struggle to adapt to the fast-evolving landscape, increasing tensions and misalignment within the network (Haarhaus & Liening, 2020). Consequently, the interaction between environmental uncertainty and collaborative innovation networks indicates that unpredictability not only diminishes the effectiveness of these alliances but also undermines their capacity to advance AI strategies, calling for a reevaluation of collaborative approaches in volatile contexts.

Based on these observations, this study presents the following hypotheses:

H3a. *Environmental uncertainty will negatively moderate the relationship between internal innovation resources and AI orientation.*

H3b. *Environmental uncertainty will negatively moderate the relationship between collaborative innovation networks and AI orientation.*

2.5. Digital culture as a moderator

Organizational culture plays a pivotal role in determining organizational effectiveness, competitive positioning, strategic alignment, and adaptability in response to changing environments (Grover et al., 2022). The alignment of organizational culture with an innovation-centric culture is crucial for enterprises to facilitate digital transformation and gain a competitive edge (Gurbaxani & Dunkle, 2019). Achieving this transformation requires a rethinking of the organization's vision, strategy, structure, processes, capabilities, and culture to align with the demands of the digital business landscape. Digital culture, defined as the fusion of information technology and business strategy, gives rise to new organizational practices, strategies, and structures (Guy, 2019). Many organizations are now selectively incorporating elements of digital culture while preserving their core traditional competencies (Westerman et al., 2019).

Digital culture encompasses an emerging set of values, practices, and expectations that guide organizational behaviors and interactions in our hyper-connected society (Proksch et al., 2021). These values, practices, and beliefs foster an organizational environment that views digital technologies as opportunities rather than threats, creating a climate conducive to continuous learning, innovation, and technological adaptability (Andriopoulos & Lewis, 2009; Westerman et al., 2014).

Strategically, digital culture aligns organizational culture with the goals of digital transformation, emphasizing agility, flexibility, and responsiveness (Nambisan et al., 2017). It also signifies a shift from a product-centric to a user-centric focus in digital innovation, highlighting the importance of customer engagement and experience (Bharadwaj et al., 2013). Westerman et al. (2019) identified four key values inherent to digital culture: influence, speed, openness, and autonomy.

Our discussion posits that digital culture serves as a lever that amplifies the effectiveness of corporate resources in shaping AI strategies (Proksch et al., 2021). Concerning internal innovation resources, a vibrant digital culture enhances an organization's ability to understand, assess, and implement emerging technologies like AI. This amplifies the utility of internal innovation resources and ensures their alignment with the right technological trajectories (Karahanna et al., 2006). A core characteristic of digital culture is its capacity to break down organizational silos, promote cross-functional collaboration, and foster a culture of experimentation (Nambisan et al., 2019). Such an environment is conducive to the effective utilization of internal innovation assets. In the context of AI orientation, organizations immersed in digital culture are well-prepared, both in terms of resources and cultural predispositions, to seamlessly integrate AI into their operational frameworks (Davenport & Ronanki, 2018). Moreover, the inherently risk-tolerant and experimental nature of digital culture synergizes with internal innovation resources to drive a proactive, comprehensive AI strategy that spans from ideation to execution (Grover et al., 2022).

From an external collaboration perspective, digital culture enhances an organization's ability to cultivate stronger inter-organizational linkages and collaborations (Martínez-Caro et al., 2020). Digitally mature organizations leverage their infrastructure and cultural openness to derive value from, and contribute to, external innovation networks (Nambisan et al., 2019; Si et al., 2023). The convergence of a receptive digital culture and collaborative innovation networks positions organizations to effectively adopt and integrate AI solutions (Fountain et al., 2019). Consequently, this article presents the following hypotheses:

H4a. *Digital culture will positively moderate the relationship between internal innovation resources and AI orientation.*

H4b. *Digital culture will positively moderate the relationship between collaborative innovation networks and AI orientation.*

2.6. Executives' not-invented-here syndrome as a moderator

The establishment of structured external knowledge interactions is of paramount importance for enterprises striving to devise enduring strategies that confer a sustainable competitive advantage (Foss et al., 2013; Laursen & Salter, 2006). The cognitive phenomenon known as the "Not-Invented-Here Syndrome" (NIHS) was initially conceptualized by Katz and Allen (1982). Through extensive empirical research, they demonstrated that while performance initially improves and stabilizes over 1.5 years, a decline becomes evident by the fifth year. This decline is largely attributed to a significant reduction in communication rates among team members and critical external information sources. NIHS, rooted in internal research and development and innovation, involves the reluctance and rejection of knowledge originating from different professional domains, distinct organizational units, or geographically distant sources (Antons & Piller, 2015; Arias-Perez & Velez-Jaramillo, 2022a; Hannen et al., 2019). Initially focused on an individual or organizational disposition towards external knowledge, this concept has been extended to encompass the acceptance or rejection of emerging technologies. It defines the psychological inclination of executives to hinder or dismiss external technological inputs (Arias-Perez & Velez-Jaramillo, 2022a).

When NIHS permeates key decision-making ranks, its effects extend across project teams, departments, and entire organizations, fostering an insular mindset (Hussinger & Wastyn, 2016). Executives' NIHS can manifest both unconsciously and intentionally. This involves, firstly, the

need to define, express, and protect one's self-identity, self-concept, and values while nurturing interpersonal relationships. Secondly, it includes confirmation bias, which perpetuates a focus solely on perspectives aligning with personal viewpoints and preconceptions regarding external digital technologies (Hannen et al., 2019). Thirdly, personal opportunism comes into play, as executives have vested interests in reaping rewards while avoiding risks. Novel ideas from external sources often evoke apprehension among employees, leading to their under-valuation in favor of championing internal projects and competencies. This can result in a form of "knowledge localism" within organizations inclined towards NIHS (Burcharth et al., 2014).

NIHS among executives, as observed in AI strategy formulation, presents a dual nature. As emphasized by Arias-Perez and Velez-Jaramillo (2022a), the inception of a corporate AI orientation hinges on executives recognizing and incorporating AI. Those affected by NIHS tend to favor internal innovation, perceiving in-house solutions as superior to external AI interventions. There are two primary manifestations of this: First, such executives exhibit a preference for endorsing in-house developed solutions, resulting in a heightened reliance on internal innovation reservoirs for AI (Amann et al., 2022). Consequently, there is an increase in financial, human, and attentional resources directed towards AI initiatives grounded in the organization's inherent knowledge and technological expertise. Second, NIHS-afflicted executives leverage internal innovation not only as a technical asset but also as a tool for social cohesion, reinforcing the collective organizational identity (Antons et al., 2017). Hence, we propose that:

H5a. *Executives' not-invented-here syndrome will positively moderate the relationship between internal innovation resources and AI orientation.*

However, the counterpoint is that NIHS might impede the expansion of AI strategies through collaborative innovation networks. Firstly, NIHS can influence employees' information search and assimilation processes, introducing biases where content is evaluated more based on its origin than its intrinsic value. This resulting "attitudinal selectivity effect" fosters skewed attention, interpretation, and retention patterns, potentially sidelining valuable external insights (Burcharth et al., 2014). Strategically, this inclination might lead to a preference for internal solutions and an unintentional sidelining of external perspectives, creating an inherent divide between internal and external innovation (Lichtenthaler & Ernst, 2006; Lucas & Goh, 2009).

Secondly, if firms fall victim to NIHS, they may have reservations about adopting external AI solutions, especially in situations where their in-house technical resources are not robust. NIHS could paralyze executives, making them hesitant to pursue external collaborations (Lichtenthaler, 2020). Such hesitation could delay the adoption or infusion of AI technologies, putting the firm's adaptability and agility in the AI landscape at risk and compromising their competitive edge (Arias-Perez & Velez-Jaramillo, 2022a).

Thirdly, the consequences of NIHS can further fragment a firm's ability to incorporate external data, limiting their knowledge base and distorting decision-making (Burcharth & Fosfuri, 2015). Executives influenced by NIHS may overlook potential partnerships or undervalue potential collaborations, limiting access to advanced AI solutions (Hannen et al., 2019). If there is overconfidence in their internal resources, they may be oblivious to the merits of external collaboration, missing out on fresh perspectives and inhibiting innovation (Lee et al., 2022). Based on this discussion, this research proposes the following hypotheses:

H5b. *Executives' not-invented-here syndrome will negatively moderate the relationship between collaborative innovation networks and AI orientation.*

A research model was developed to test the aforementioned hypotheses, with the objective of examining the direct influence of internal innovation resources and collaborative innovation networks on artificial intelligence orientation, as well as the moderating effects of environmental uncertainty, digital culture, and executives' not-invented-here syndrome. The proposed research model is illustrated in Fig. 1.

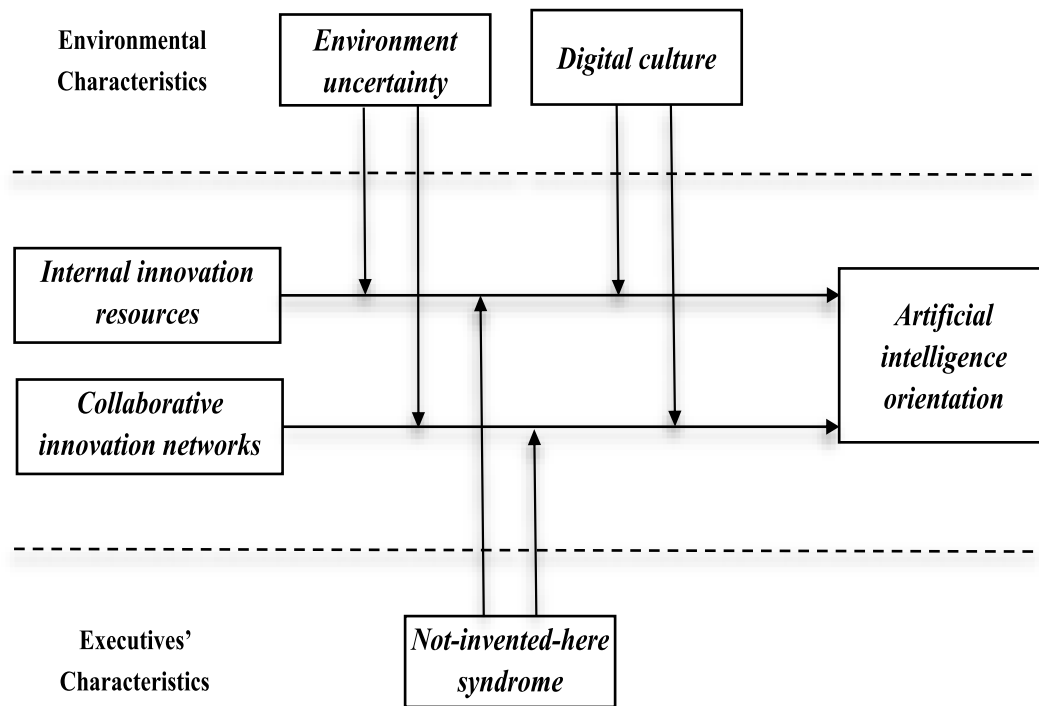


Fig. 1. Research model of innovative resources driven artificial intelligence orientation.

3. Research method

3.1. Sample and data collection

The study gathered survey data using the Prolific platform, a tool well-regarded in management scholarship for its effectiveness in administering questionnaires (Jeesha & Purani, 2021; Kossyva et al., 2023). Prolific provides extensive coverage of the global working population (Peer et al., 2017). Our survey was conducted among executives from diverse countries. Among the 800 questionnaires distributed via the platform, several credibility checks were implemented to ensure data integrity. Initially, we pre-screened senior managers as our target respondents and included questions to confirm their positions, verifying that the majority were indeed senior enterprise managers. Secondly, we cross-referenced respondents' company names or websites with geographical indicators to authenticate their organizations. Thirdly, open-ended questions exploring the organizations' current AI strategies and future directions further reinforced response credibility.

A meticulous evaluation of these submissions confirms the authenticity of the collected data. Ultimately, 717 responses were deemed valid, representing 89.625% of the initial pool. Comprehensive details on respondent and organizational characteristics are presented in Table 1.

3.2. Variables and measures

In the current study, the measurement items for the relevant variables were derived and structured through a rigorous analysis and synthesis of prior scholarly works. These items were assessed using a 7-point Likert scale, ranging from 1 (strongly disagree) to 7 (strongly agree).

AI Orientation. Prior research on AI orientation predominantly employed word frequency analyses of AI mentions within annual reports of publicly traded companies to gauge AI orientation (Arias-Pérez & Vélez-Jaramillo, 2022b; Li et al., 2021, 2023). In this study, drawing upon the aforementioned literature, a 13-item scale was devised to assess AI orientation. This scale evaluated the extent to which

companies have integrated AI technology, innovated using AI, formulated relevant strategic directives, articulated clear visions, refined organizational structures, determined AI investment strategies, and merged AI with overarching business strategies to harness the potential value AI imparts to business operations. A sample item was "our company has a clear vision regarding how AI technologies (such as Robotics, Image Recognition, Language Recognition, Deep Learning, Natural Language Processing) contribute to business value". The reliability coefficient was 0.951.

Internal Innovation Resources. Drawing on the innovation resources scale developed by Halme and Korpela (2014), this study employed a four-item questionnaire to measure the level of a firm's internal innovation resources. These questionnaire items are designed to evaluate whether the firm possesses a more comprehensive array of innovative components compared to its industry peers. A sample item was "compared to others in the same industry, our company has good profit ability and can provide sufficient financial support for innovation activities". The reliability coefficient was 0.830.

Collaborative Innovation Networks. Anchored in the research conducted by Najafi-Tavani et al. (2018), this study employs a four-item scale to quantify the scope and depth of collaborative innovation networks. Respondents were prompted to delineate the degree to which their organizations foster external collaborations spanning suppliers, consumers, competitors, academic institutions, and R&D entities. A sample item was "our company collaborates with suppliers for innovation". The reliability coefficient was 0.717.

Environmental Uncertainty. Environmental uncertainty signals the unpredictability companies face across various domains such as markets, technology, and supply chains. Drawing upon Liao et al. (2011), this study gauged environmental uncertainty through three question items that illuminate the breadth and diversity of customers, suppliers, and competitors. A sample item was "Our company faces a diversity of customer needs". The reliability coefficient was 0.804.

Digital Culture. A robust digital culture champions the fostering of collaborative workspaces, galvanizes creativity and innovation, and propels challenge, initiative, and sustained enhancements via cohesive digital strategies (Kane et al., 2015). Informed by the research of

Table 1
Respondents and surveyed companies' information summary.

Characteristics	Types	Number	Percentage (%)
Gender	Male	438	61.1
	Female	279	38.9
Age	≤25 years old	51	7.1
	26–35 years old	209	29.1
	36–45 years old	192	26.8
	>46 years old	265	37
Education	Junior college and below	141	19.7
	Bachelor's degree	343	47.8
	Master's degree	193	26.9
	PhD or post-doctoral	40	5.6
Industry	Basic Materials	21	2.9
	Consumer Goods	164	22.9
	Energy	14	2
	Finance	92	12.8
	Medical and Healthcare	60	8.4
	Industry	88	12.3
	Information Technology	197	27.5
	Telecommunication Services	30	4.2
	Utilities	16	2.2
	Real Estate	35	4.9
	<3 years	62	10.3
	3–5 years	75	10.5
Established years	5–10 years	125	17.4
	>10 years	455	63.5
Number of employees	≤100	353	49.2
	101–300	99	13.8
	301–500	58	8.1
	501–1000	59	8.2
	≥1001	148	20.6
Use of AI technologies	Machine learning: Deep learning, unsupervised, supervised	276	38.49
	Natural language processing(NLP)	339	47.28
	Expert systems	146	20.36
	Speech: Speech to text, text to speech	335	46.72
	Vision: Image recognition, machine vision	199	27.75
	Planning	266	37.10
	High intelligence robotics	58	8.09
	Others	85	11.85

Martínez-Caro et al. (2020), this study assessed an organization's digital culture using four question items, each signifying the organization's propensity towards embracing technological shifts, championing digital innovation, and adopting collaborative digital strategies. A sample item was “our company collaborate functionally in the initiatives for the innovation and digital transformation”. The reliability coefficient was 0.794.

Executive's Not-invented-here Syndrome. The NIHS paradigm encapsulates the degree to which corporate leaders rebuff external knowledge and technological inputs. Based on insights from Arias-Perez and Velez-Jaramillo (2022a) and Burcharth et al. (2014), this research evaluated executives' inclination towards NIHS utilizing three reverse-scored question items. A sample item was “I regard the application of external knowledge as valuable as the application of knowledge generated inside”. The reliability coefficient was 0.807.

Control Variables. This investigation employs a robust control strategy, incorporating both respondent-level and firm-level variables, to bolster the rigor of the empirical analysis. At the respondent level, demographic characteristics—including gender, age, and education level—are controlled to mitigate potential individual biases that could distort perceptions or behaviors related to the focal constructs, specifically the assimilation or appraisal of AI technologies. At the organizational level, structural attributes such as firms' age and employee count are included as control variables to account for inherent organizational differences that may influence a firm's innovative propensity and openness to emerging technologies (Jain & Mitchell, 2022). Table 2 provides a detailed enumeration of constructs employed in this study,

Table 2
Constructs and definitions.

Construct	Definition	Source(s)
Artificial intelligence orientation	Artificial intelligence orientation represents an important paradigm shift within enterprises, indicating a strategic commitment to harnessing AI technology for transformative purposes, including the adoption and implementation of AI solutions for intelligent transformation.	Arias-Pérez & Vélez-Jaramillo, 2022b; Li et al., 2021; Li et al., 2023
Internal innovation resources	Internal innovation resources encompass innovative human resources, innovative financial resources, as well as technological and informational resources.	Halme and Korpela (2014)
Collaborative innovation networks	Collaborative innovation networks represent an enterprise's engagement with a diverse range of partners, including suppliers, customers, competitors, and research organizations, with the goal of integrating complementary knowledge and competencies.	Najafi-Tavani et al. (2018)
Environmental uncertainty	Environmental uncertainty refers to the degree to which business decision-makers find it difficult to anticipate or accurately predict external changes, including fluctuations in competitor actions, sociocultural trends, regulatory policies, and technological advancements.	Liao et al. (2011)
Digital culture	Digital culture, defined as the fusion of information technology and business strategy, gives rise to new organizational practices, strategies, and structures.	Martínez-Caro et al. (2020)
Not-invented-here syndrome	Not-Invented-Here Syndrome defines the psychological inclination of executives to hinder or dismiss external technological inputs.	Arias-Perez & Velez-Jaramillo, 2022a; Burcharth et al., 2014

complete with their definitions and measurement sources. This tabulation is instrumental in elucidating the conceptual foundations of our research framework.

The reliability coefficients of studies variables, as show in Table 3, ranged from 0.717 to 0.951. Furthermore, the composite reliability (CR) values surpassed 0.7, well above the recommended minimum threshold of 0.5, denoting excellent construct reliability for the scale (Hair et al., 2019). Thirdly, all variables demonstrated an average variance extracted (AVE) greater than 0.5, signifying the effective reflection of latent variables by indicator variables. This observation attests to the scale's good reliability, validity, and supports the convergent validity of the construct measure (Henseler et al., 2016).

3.3. Common method bias

To mitigate potential common method bias in this study, a combination of procedural and measurement controls was implemented. Procedurally, anonymous completion of surveys was facilitated, and question items were presented in a mixed order to deter systematic response patterns (Palacios-Manzano et al., 2021). In terms of measurement controls: Initially, the potential for endogeneity in the scale was assessed using Variance Inflation Factor (VIF) values. Consistent with Hair et al. (2017), VIF values for all variables were found to be

Table 3
Reliability and validity of variables.

Variables	Items	Factor loadings	α	CR	AVE
AI Orientation	AI_Or 1	0.782	0.951	0.954	0.63
	AI_Or 2	0.874			
	AI_Or 3	0.82			
	AI_Or 4	0.819			
	AI_Or 5	0.744			
	AI_Or 6	0.787			
	AI_Or 7	0.845			
	AI_Or 8	0.61			
	AI_Or 9	0.801			
	AI_Or 10	0.864			
	AI_Or 11	0.853			
	AI_Or 12	0.779			
	AI_Or 13	0.741			
Internal innovation resources	IIR_1	0.681	0.830	0.852	0.664
	IIR_2	0.858			
	IIR_3	0.86			
	IIR_4	0.846			
Collaborative innovation networks	CIN_1	0.751	0.717	0.723	0.538
	CIN_2	0.73			
	CIN_3	0.702			
	CIN_4	0.751			
Not-invented-here syndrome	NIHS_1	0.845	0.807	0.830	0.716
	NIHS_2	0.852			
	NIHS_3	0.842			
Digital culture	DC_1	0.781	0.794	0.797	0.618
	DC_2	0.817			
	DC_3	0.784			
	DC_4	0.761			
Environmental uncertainty	EU_1	0.851	0.804	0.805	0.718
	EU_2	0.857			
	EU_3	0.835			

below the threshold of 3, suggesting an absence of endogeneity concerns. Subsequently, an Unmeasured Latent Common Method Factor (ULCMF) approach was adopted to assess common method variance (Podsakoff et al., 2003). Confirmatory factor analysis (CFA) indicated that the fit of a six-factor model was appreciably superior to that of a one-factor model, evidenced by statistical metrics: $\chi^2/df = 2.856$, CFI = 0.935, TLI = 0.923, RMSEA = 0.049. Furthermore, a comparison between the six-factor model's fit and the ULCMF measurement model exhibited marginal differences in fit indices ($\Delta\chi^2/df = 0.608$, $\Delta CFI = 0.031$, $\Delta TLI = 0.033$, $\Delta RMSEA = 0.012$). Given these diagnostics, the study confidently concludes that the common method bias is not a significant concern in the measurements. An elaborative tabulation of these diagnostic results is presented in Table 4 under 'Common Method Bias Analysis.'

4. Analysis results

4.1. Correlation analysis

The means, standard deviations, and Pearson correlation coefficients of studied variables are showed in Table 5.

Table 4
Common method bias analysis results.

	χ^2	df	CFI	TLI	RMSEA
Single-factor model	2085.464	434	0.861	0.841	0.071
Six-factor model	1196.868	419	0.935	0.923	0.049
ULCMF	797.151	388	0.966	0.956	0.037

4.2. Hypothesis testing results

This study conducted regression analyses to examine the previously presented hypotheses, with the results summarized in Table 6. Model 1 explored the influence of control variables on artificial intelligence orientation, whereas Model 2 investigated how internal innovation resources and collaborative innovation networks shape enterprises' AI strategies.

The findings in Model 2 revealed that the path coefficient for internal innovation resources on AI orientation was 0.682 ($t = 23.142$, $p < 0.001$), confirming H1. Additionally, the regression coefficient for the collaborative innovation network on AI strategy was 0.129 ($t = 4.393$), confirming H2.

Regarding moderating effects, Model 3 demonstrated the moderating impact of environmental uncertainty on the relationship between internal innovation resources and AI orientation. The results indicated that environmental uncertainty negatively moderates this relationship ($\beta = -0.314$, $t = 2.054$), confirming H3a.

Model 4 showed that environmental uncertainty also negatively moderates the relationship between the collaborative innovation network and AI orientation, with a regression coefficient of -0.423 ($t = 2.159$, $p < 0.05$), thereby establishing H3b.

Models 5 and 6 illuminated the moderating role of digital culture. Model 5 suggested that digital culture positively moderates both the relationship between internal innovation resources and AI orientation ($\beta = 0.415$, $t = 2.994$, $p < 0.01$) and the relationship between the collaborative innovation network and AI orientation ($\beta = 0.377$, $t = 2.398$, $p < 0.05$), confirming H4a and H4b, respectively.

Models 7 and 8 explored the moderating effect of executives' Not-Invented-Here Syndrome (NIHS). The results revealed that the regression coefficient for executives' NIHS between internal innovation resources and AI orientation was 0.258 ($t = 2.359$, $p < 0.05$), confirming H5a. However, the moderating effect of NIHS between the collaborative innovation network and AI orientation was not significant ($\beta = -0.139$, $t = 0.994$), leading to the non-confirmation of H5b.

4.3. Robustness check

Beyond the control variables of respondent information and firm-specific data, it is crucial to consider external factors that could influence the formulation of AI strategies. These include the economic conditions of the firms' home countries, the level of digitization in those countries, and the specific industries in which the firms operate, as these external elements can exert substantial impacts on a company's AI strategy development.

Firstly, the economic condition of the country of operation plays a pivotal role. Robust economies empower companies to allocate substantial resources to research and development, thereby enhancing their capacity to adopt AI technologies for competitive advantage. Conversely, enterprises in weaker economic environments may need to judiciously allocate resources to ensure the success of their AI strategies.

Secondly, the penetration of AI within a nation and its industries has significant implications. High AI adoption facilitates seamless integration and operationalization of AI technologies, given the established infrastructure and skilled human capital. In contrast, regions with nascent AI development may face extended timelines and increased resource allocations for effective AI assimilation.

Lastly, distinct industries demonstrate varying dependencies on AI adoption. For example, the technology and finance sectors often embrace AI early due to its integral role in data analytics and process automation. However, traditional manufacturing industries may require extensive reform to fully harness AI's potential. Therefore, comprehensive research must encompass these critical factors to provide a nuanced understanding of the determinants influencing a company's AI strategy.

Hence, this study employed two robustness testing methods to

Table 5

Descriptive statistics and correlations.

Variables	Mean	SD	1	2	3	4	5	6	7	8	9	10
1. AI Orientation	4.57	1.29										
2. Internal innovative resources	4.34	1.30	0.764**									
3. Collaborative innovation networks	4.38	1.24	0.548**	0.611**								
4. Digital culture	5.06	1.06	0.726**	0.664**	0.558**							
5. Environmental uncertainty	4.77	0.93	0.233**	0.194**	0.231**	0.278**						
6. Not-invented-here syndrome	2.56	0.65	-0.232**	-0.19**	-0.211**	-0.31**	-0.197**					
7. Gender	1.39	0.49	-0.09*	-0.082*	0	-0.078*	-0.053	-0.04				
8. Age	2.94	0.97	-0.133**	-0.127**	-0.165**	-0.10**	-0.048	0.054	-0.056			
9. Education	2.18	0.81	0.104**	0.104**	0.101**	0.083*	0.101**	-0.047	-0.076*	0.036		
10. Established years	4.34	1.03	-0.063	0.004	-0.022	-0.024	0.028	0.025	-0.044	0.266**	0.116**	
11. Number of employees	2.37	1.62	0.107**	0.187**	0.165**	0.073	0.073	-0.039	-0.04	0.004	0.152**	0.338**

NOTE. N = 717. **, p < 0.01; *, p < 0.05.**Table 6**

Regression analysis results.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Intercept								
Gender	-0.088* (2.476)	-0.036(1.529)	-0.025(1.060)	-0.083** (2.78)	-0.025(1.194)	-0.046(1.899)	-0.032(1.395)	-0.094** (3.149)
Age	-0.119** (3.211)	-0.013(0.544)	-0.024(0.996)	-0.039(1.246)	-0.018(0.822)	-0.029(1.154)	-0.018(0.762)	-0.037(1.171)
Education	0.094***(2.611)	0.029(1.225)	0.027(1.126)	0.039(1.274)	0.027(1.297)	0.035(1.436)	0.027(1.146)	0.046(1.506)
Established time	-0.086* (2.194)	-0.055*(2.136)	-0.059*(2.288)	-0.064(1.924)	-0.057* (2.496)	-0.059* (2.213)	-0.059*(2.303)	-0.059(1.782)
Number of employees	0.199***(3.114)	-0.029(1.159)	-0.023(0.929)	0.031(0.957)	-0.007(0.313)	0.039(1.491)	-0.022(0.859)	0.028(0.856)
Direct effect								
IIR		0.682*** (23.142)	0.987*** (8.158)		0.222*(2.293)		0.533*** (5.796)	
CIN		0.129****(4.393)		0.825*** (5.500)		-0.059(0.538)		0.619*** (5.276)
EU			0.237*(3.084)	0.314***(3.095)				
DC					0.218** (3.461)	0.439****(5.85)		
NIHS							-0.274** (3.352)	-0.019(0.177)
Moderating effect								
IIR*EU			-0.314** (2.054)					
CIN*EU				-0.423* (2.159)				
IIR*DC					0.415** (2.994)			
CIN*DC						0.377*(2.398)		
IIR*NIHS							0.258*(2.359)	
CIN*NIHS								-0.139(0.994)
R ²	0.052	0.600	0.599	0.331	0.677	0.568	0.601	0.331
Adj. R ²	0.045	0.597	0.595	0.323	0.674	0.564	0.596	0.324
F-value	8.220***	161.562***	140.698***	46.415***	197.091***	123.644***	141.382***	46.548***

NOTE. N = 717. IIR: Internal innovative resources; CIN: Collaborative innovation networks; DC: Digital culture; EU: Environmental uncertainty; NIHS: Not-invented-here syndrome. ***, p < 0.001; **, p < 0.01; *, p < 0.05.

ensure a comprehensive analysis: incorporating previously omitted variables and categorizing enterprises for further hypothesis testing. First of all, to enhance the rigor of our analysis and account for potential effects from country-level economic metrics and the broader AI landscape, we drew from seminal databases such as the “World Economic Outlook” by the International Monetary Fund (IMF) (Chen & Ranciere, 2019) and the Global Artificial Intelligence Index by Tortoise (Erkkila, 2023). The IMF’s World Economic Outlook provides extensive metrics—including nominal GDP, per capita GDP, and mid-year demographic estimates—for its 193 member nations. It’s noteworthy to highlight that, as per the report, the U.S. emerges preeminent in nominal GDP metrics at \$25.46 trillion, while Luxembourg boasts the apex position in per capita GDP metrics at \$125,600. Concurrently, Tortoise’s Global Artificial Intelligence Index offers a multi-dimensional appraisal of 62 nations based on parameters such as Talent, Infrastructure, Operating

Environment, Research and Development, Government Strategy, Commercial Viability, and AI Intensity, positioning the U.S., China, and Singapore at the forefront of global standings.

In an effort to establish the robustness of our analytical framework, we juxtaposed the per capita GDP figures from the 2022 World Economic Outlook by the IMF against the Global AI Index rankings by Tortoise and further examined industry nuances. From the original 717 samples, 12 were excluded due to the absence of evaluations in the Global Artificial Intelligence Index, resulting in a refined sample size of 705. By integrating variables such as the artificial intelligence index metrics, per capita GDP, and industry – detailed in Table 7 – our outcomes corroborate the study’s postulated hypotheses. This underscores the robustness of our model in accommodating disparities in both economic stature and AI developmental stages across diverse nations.

Furthermore, to delve into the industrial disparities in the

Table 7

Robustness check results.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Intercept								
Gender	−0.096** (2.569)	−0.043(1.729)	−0.033(1.339)	−0.083** (2.599)	−0.04(1.778)	−0.06*(2.334)	−0.04(1.603)	−0.094** (2.94)
Age	−0.054(1.328)	0.01(0.382)	0.004(0.134)	−0.009(0.246)	0.009(0.358)	0.001(0.041)	0.009(0.324)	−0.009(0.269)
Education	0.093*(2.526)	0.036(1.486)	0.033(1.365)	0.041(1.302)	0.035(1.601)	0.042(1.657)	0.034(1.388)	0.05(1.585)
Established time	−0.069(1.724)	−0.055*(2.052)	−0.058* (2.175)	−0.057(1.666)	−0.06* (2.493)	−0.059* (2.129)	−0.058(2.183)	−0.052(1.52)
Number of employees	0.106***(2.72)	−0.026(0.973)	−0.021(0.801)	0.027(0.816)	−0.004 (0.182)	0.039(1.418)	−0.019(0.745)	0.023(0.686)
Industry	0.132*** (3.578)	0.023(0.937)	0.029(1.177)	0.105*** (3.346)	−0.003 (0.116)	0.026(0.997)	0.026(1.047)	0.096***(3.046)
Talent	−0.37(1.043)	0.161(0.682)	0.162(0.686)	−0.197(0.652)	0.242(1.142)	0.107(0.436)	0.203(0.861)	−0.181(0.597)
Infrastructure	−0.409(1.369)	−0.313(1.581)	−0.336(1.696)	−0.345(1.353)	−0.348 (1.953)	−0.396(1.921)	−0.325(1.646)	−0.358(1.405)
Operating Environment	0.028(0.513)	−0.01(0.262)	0.007(0.184)	−0.025(0.535)	0.012(0.358)	0.013(0.344)	−0.002(0.057)	−0.037(0.793)
Research	0.793(1.172)	0.007(0.015)	0.089(0.199)	0.166(0.287)	−0.106 (0.264)	−0.066(0.14)	0.004(0.01)	0.114(0.196)
Development	0.697(1.496)	0.464(1.503)	0.479(1.552)	0.606(1.525)	0.398(1.432)	0.469(1.457)	0.462(1.499)	0.535(1.343)
Government Strategy	0.031(0.232)	0.041(0.467)	0.026(0.292)	0.033(0.294)	0.002(0.022)	−0.01(0.11)	0.054(0.607)	0.074(0.65)
Commercial	−0.492(0.934)	−0.206(0.590)	−0.286(0.821)	−0.141(0.313)	−0.123 (0.393)	−0.071(0.195)	−0.223(0.640)	−0.009(0.02)
Intensity	−0.23(1.391)	−0.213(1.93)	−0.25*(2.279)	−0.046(0.328)	−0.189 (1.912)	−0.1(0.875)	−0.267*(2.446)	−0.058(0.406)
Per capita GDP	−0.091(0.76)	0.04(0.504)	0.077(0.966)	−0.055(0.535)	0.089(1.25)	0.047(0.571)	0.063(0.797)	−0.075(0.734)
Direct effect								
IIR		0.674*** (21.786)	0.993*** (7.956)		0.258** (2.571)		0.511*** (5.371)	
CIN		0.116*** (3.756)		0.831*** (5.406)		−0.016(0.143)		0.593*** (4.898)
EU			0.248*(3.105)	0.335***(3.226)				
DC					0.24***(3.678)	0.469*** (6.038)		
NIHS							−0.276*** (−3.27)	−0.014(0.121)
Moderating effect								
IIR*EU			−0.345** (2.183)					
CIN*EU				−0.464* (2.307)				
IIR*DC					0.350*(2.43)			
CIN*DC						0.301*(2.078)		
IIR*NIHS							0.265*(2.345)	
CIN*NIHS								−0.134(0.931)
R ²	0.116	0.613	0.614	0.361	0.688	0.581	0.615	0.359
Adj. R ²	0.096	0.604	0.604	0.344	0.680	0.57	0.605	0.342
F-value	5.996***	63.979***	60.600***	21.593***	83.919***	52.846***	60.902***	21.267***

NOTE. N = 705. IIR: Internal innovative resources; CIN: Collaborative innovation networks; DC: Digital culture; EU: Environmental uncertainty; NIHS: Not-invented-here syndrome. ***, $p < 0.001$; **, $p < 0.01$; *, $p < 0.05$.

formulation of AI orientations, we categorized firms into tech-based and non-tech groups, relying on the survey respondents' classifications. Subsequent to this categorization, we gathered data from a sample comprising 410 tech-based firms and 307 non-tech firms. A comparative analysis of their AI orientation levels unveiled that tech-based firms exhibited a mean AI orientation score of 4.75, whereas non-tech firms scored 4.34. Hypothesis testing conducted for both categories yielded statistically significant results, albeit without discernible differences between them.

5. Discussion

In this study, we employed an integrative framework that merges the resource-based view and strategic choice theory to explore the determinants and contextual constraints influencing firms in the formulation of artificial intelligence strategies. Using a comprehensive dataset from 717 questionnaires, our empirical analysis reveals the crucial roles of internal innovation resources and collaborative innovation networks in facilitating firms' adoption of an AI orientation. Moreover, our findings delve into the moderating effects of environmental uncertainty,

digital culture, and the executive's "not-invented-here" syndrome on these relationships. Collectively, these insights provide a nuanced understanding of the intricate interplay of factors shaping AI strategy formulation within firms, emphasizing the necessity for organizations to navigate both internal and external influences to effectively harness AI's potential in today's dynamic business environment.

5.1. Theoretical contributions

This study significantly enhances our scholarly understanding of AI strategy formulation within firms. By furnishing empirical evidence and insights into both the primary impacts of internal and external resources and the moderating influences of environmental uncertainty, digital culture, and executive NIHS on AI orientation, this research presents a nuanced and multifaceted theoretical contribution to the field.

Firstly, we conducted a thorough analysis of the concept of artificial intelligence orientation and developed a measurement scale, deepening the theoretical landscape and strategic dimensions of AI. By exploring the core connotations of AI orientation, we clarified its role in shaping enterprise strategy, addressing aspects such as IT and digital

capabilities, innovation, and competitive positioning (Alter, 2022; Muhlroth & Grottko, 2022). Meanwhile, we developed an innovative measurement scale to systematically assess AI orientation, bridging a notable gap in the literature. Unlike prior studies relying on secondary data from listed companies (Li et al., 2021, 2023), our scale captures critical dimensions such as R&D investment, technology development, value creation, and cultural integration, offering a more precise and comprehensive tool for evaluating AI strategies. This refined approach deepens the theoretical landscape by broadening the strategic manifestations of AI within organizations and expanding its theoretical boundaries. It contributes to a nuanced understanding of AI's role in driving innovation and competitive advantage, enriching the discourse on AI's integration into business strategies (Mikalef & Gupta, 2021).

Secondly, this study fills a significant research gap by investigating the determinants of artificial intelligence orientation, specifically innovation resources and contextual constraints such as environmental uncertainty, digital culture, and the not-invented-here syndrome (Fountain et al., 2019). While AI strategies have garnered increasing attention in recent years (Li et al., 2023; Li et al., 2021), systematic investigation into the factors driving their development remains limited. Our theoretical model demonstrates that internal innovation resources and collaborative innovation networks are essential for AI strategy formulation. Additionally, we examine the boundary conditions under which these resources shape AI strategies, specifically focusing on environmental and executive characteristics. We find that environmental uncertainty negatively moderates the relationship, while digital culture positively moderates it. This study marks the inaugural scholarly endeavor to incorporate the not-invented-here syndrome into AI orientation research. Departing from the conventional focus in management literature on the negative effects of NIHS—where excessive prevalence among enterprise managers can lead to impediments, failure to share information, sabotage, or other deviant behaviors, as well as stress, frustration, and other psychological consequences (Antons & Piller, 2015)—our innovative approach demonstrates that NIHS has differential impacts: it bolsters the influence of internal innovation resources on AI strategy formulation while simultaneously diminishing the effectiveness of collaborative innovation networks. This refined interpretation not only enriches strategic management discourse but also clarifies how executive biases influence resource allocation within the sphere of AI-driven innovation. Our pioneering findings delineate new boundary conditions and provide essential insights into AI strategy development, thereby addressing a substantial void in the existing literature (Li et al., 2021).

Thirdly, this study integrates the resource-based view and strategic choice theory into a unified theoretical framework, enhancing its theoretical applicability in comprehending innovation strategy in the digital era. As companies navigate digital transformation, they encounter pressures not only from technological advancements—such as the rapidly evolving AI landscape and the uncertainties surrounding generative AI technologies—but also from the need for flexible strategic adjustments in dynamic market conditions. The integration of resource-based view, which emphasizes the role of enterprise resources in shaping competitive advantages (Barney, 1991; Tiwari et al., 2024), with strategic choice theory, which highlights dynamic strategic decision-making in fluctuating environments (Child, 1997; Montanari, 1979), offers a robust framework. This approach underscores the importance of leveraging internal resources while adapting strategies in response to external environmental shifts, enabling firms to overcome constraints and secure competitive advantages (Helfat et al., 2023). By broadening theoretical perspectives, this integration provides valuable insights for future research on corporate innovation strategies amidst AI-driven technological disruptions.

5.2. Managerial implications

This study highlights several practical implications for the

development of AI strategies in enterprises, aligning with contemporary management paradigms. Firstly, enterprises are increasingly embracing AI as a strategic imperative, investing in technologies to strengthen their competitive advantage. AI is no longer just a tool but a critical element of enterprise strategy. Effective AI strategy formulation and implementation are essential for gaining a competitive edge in today's markets. Integrating automation, predictive analytics, and intelligent decision-making systems boosts productivity and reduces inefficiencies, optimizing operations. Generative AI, capable of creating text, images, and designs, presents new avenues for innovation. Companies can use it to personalize products, accelerate design processes, and enhance creativity, reducing time-to-market. Additionally, deep learning and big data analytics enable firms to identify market trends, understand customer needs, and drive innovation. As AI rapidly evolves, managers must integrate it into strategic planning, ensure comprehensive understanding of its capabilities, invest in employee training, and prioritize ethical considerations—especially in the responsible use of Generative AI. Moreover, organizations must cultivate strategic agility and adopt a proactive approach to ensure the continued relevance of their operational frameworks. This entails the continuous refinement of AI strategies, strategic investments in scalable technological infrastructures, and the cultivation of an organizational culture that prioritizes innovation and continuous learning. Such an approach enables enterprises to maintain their competitive advantage by rapidly assimilating novel capabilities and insights, thereby ensuring alignment with evolving industry standards and technological advancements.

Secondly, companies with internal innovation resources and collaborative innovation networks are advantageously positioned for the effective formulation and implementation of AI strategies. In the rapidly evolving contemporary business landscape, internal innovation resources—spanning talent, technology, processes, and culture—constitute the cornerstone for organizations tackling AI challenges. The harmonious integration of AI-expert internal teams, adaptable organizational structures, and an innovative culture fortifies the support for AI strategies. Simultaneously, the establishment of a collaborative innovation network becomes critically important. By fostering a tightly interlinked network with external partners, suppliers, and research institutes, organizations not only gain early access to pioneering technological advancements but also catalyze innovation and upgrading across the entire industrial ecosystem. As AI strategy development progresses, management must scrutinize the enterprise's internal innovation resources, constructing a specialized internal team with heightened AI capabilities. In parallel, proactive cultivation of a collaborative innovation network with external entities is vital for effectively navigating industry transformations. Importantly, the iterative upgrading of internal innovation capabilities should be synchronized with the strategic expansion of external collaborative networks, creating a synergistic effect that enhances the timeliness and superiority of AI strategy implementation. Furthermore, establishing a dual-driven innovation system that integrates internal R&D breakthroughs with external ecosystem co-creation will significantly improve the organization's strategic responsiveness and technological leadership in the rapidly evolving AI landscape.

Thirdly, the formulation of artificial intelligence strategies mandates that companies cultivate decision-makers with forward-looking, open-minded perspectives, ensuring decisions align harmoniously with the internal and external environments. In the digital age, navigating the rapidly evolving AI landscape necessitates a mastery of the forefront of AI technology and its trends. The development of AI should focus on optimizing deep learning, enhancing self-learning capabilities of algorithms, and encouraging cross-domain integrations. Moreover, organizations need to deeply integrate these technological advancements into their corporate culture and operational processes, ensuring the seamless implementation and innovation of their strategies. Embracing external innovations and forging strong partnerships with all stakeholders are also essential for enhancing adaptability and responsiveness to rapid

market changes. Leaders must possess acute foresight into future trends and effectively leverage opportunities presented by digital culture. Furthermore, amid escalating uncertainty, the ability to think ahead is paramount. Leaders must also transcend the constraints of the not-invented-here syndrome, actively seeking external best practices and innovations. Moving away from traditional authoritative models fosters a collaborative environment where employees feel empowered to contribute new ideas, thereby infusing a forward-thinking ethos into the organization's digital strategies. This alignment of open-mindedness with digital strategies equips companies to navigate the continuously shifting digital landscape.

5.3. Limitations and future research

This study makes a significant theoretical contribution by thoroughly examining the nuanced effects and contextual constraints of internal innovation resources and collaborative innovation networks in driving firms toward the development of AI strategies, all within a comprehensive resource perspective.

However, several limitations should be acknowledged. First, the use of self-reported questionnaires risks common method bias, which may distort relationships between predictor and outcome variables. This bias often results from cognitive consistency, social desirability, or uniform responses, inflating correlations (Han et al., 2024). Second, regarding measurement items, the noteworthy correlations observed among AI orientation, internal innovation resources, and digital culture can be attributed to the inherent nature of AI as a digital technology. AI orientation, as highlighted in this study, emphasizes the existence of a culture within the firm that shapes AI. Simultaneously, the need for AI technology research and development, coupled with capital investment, inherently correlates with the measurement of internal innovation resources. Third, the questions related to environmental uncertainty in this study place greater emphasis on the complexity and unpredictability of the environment, somewhat differing from the conventional environmental turbulence studied by scholars (Schilke, 2014; Wang & Li, 2008; Zhang et al., 2021).

Future research should adopt a mixed-methods approach, integrating quantitative and qualitative data from diverse sources—such as objective metrics, independent evaluations, and longitudinal data—to enhance the accuracy and validity of analysis. Additionally, future research could explore the mechanisms and boundary conditions that shape the impact of AI strategies on firms' innovation performance and financial outcomes. This includes investigating how AI integration across various corporate contexts, industries, and market environments interacts with human capital, organizational structure, and culture. Such insights are crucial for practitioners and policymakers aiming to drive AI-enabled competitive advantage and economic growth, especially in an increasingly dynamic AI landscape.

CRedit authorship contribution statement

Wucheng Han: Methodology, Conceptualization. **Weijie Zhu:** Formal analysis. **Zhaoli Song:** Methodology, Data curation, Conceptualization. **Ruoyu Lu:** Writing – original draft, Funding acquisition.

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Appendix

AI Orientation

Base on Arias-Pérez and Vélez-Jaramillo (2022a); Li et al. (2023); Li et al. (2021)

AI_Or 1 Our company has a clear vision regarding how AI technologies (such as Robotics, Image Recognition, Language Recognition, Deep Learning, Natural Language Processing) contribute to business value.

AI_Or 2 Our company integrated business strategy and AI strategy.

AI_Or 3 Our company supports functional area and general management in understanding the value of AI technology investments.

AI_Or 4 Our company stays up-to-date with innovations in AI technology.

AI_Or 5 Our company is committed to experimenting with AI technologies as necessary.

AI_Or 6 Our company cultivates a culture that encourages and supports the exploration of new ways to use AI technologies.

AI_Or 7 Our company is always seeking new ways to enhance the effectiveness of AI technology use.

AI_Or 8 Decisions about investing in AI are made within our company.

AI_Or 9 Our company motivates managers to apply AI in order to solve business problems.

AI_Or 10 Our company presents a clear value proposition for AI.

AI_Or 11 Our company established a clear plan for AI adoption.

AI_Or 12 Our company communicates clearly to stakeholders about the value and strategic logic guiding our use of AI.

AI_Or 13 Our company has dedicated departments or teams focusing on AI.

Internal Innovation Resources (Halme & Korpela, 2014)

IIR_1 Compared to others in the same industry, our company has good profit ability and can provide sufficient financial support for innovation activities.

IIR_2 Compared to others in the same industry, our company has a strong R & D team whose team members effectively communicate and utilize their collective knowledge to maximize the team's output.

IIR_3 Compared to others in the same industry, our company's R & D personnel exhibited higher scientific research capabilities, consistently delivering high-quality innovation projects.

IIR_4 Compared to others in the same industry, our company possesses a more extensive technical infrastructure related to AI, including experimental equipment, technological innovation platforms, etc.

Collaborative Innovation Networks (Najafi-Tavani et al., 2018)

CIN_1 Our company collaborates with suppliers for innovation.

CIN_2 Our company collaborates with customers for innovation.

CIN_3 Our company collaborates with competitors for innovation.

CIN_4 Our company collaborates with research institutes and universities for innovation.

Not-Invented-Here Syndrome (Arias-Pérez & Vélez-Jaramillo, 2022a; Burcharth et al., 2014)

NIHS_1 I regard the application of external knowledge as valuable as the application of knowledge generated inside*.

NIHS_2 I am positive toward developing new ideas, solutions and technologies for other companies*.

NIHS_3 I favor the application of technologies from outside, which overlap to a great extent to what I already know*.

Digital Culture (Martínez-Caro et al., 2020)

DC_1 Our company collaborate functionally in the initiatives for the innovation and digital transformation.

DC_2 There is a clear orientation to digital technology changes inside the company's culture.

DC_3 The culture of digital innovation and change takes part as a natural process within the company.

DC_4 Our company shares with the staff the digital strategy, taking into consideration their suggestions.

Environmental Uncertainty (Liao et al., 2011)

EU_1 Our company faces a diversity of customer needs.

EU_2 Our company faces a variety of competition.

EU_3 Our company faces dissimilarity of supplier.

Data availability

Data will be made available on request.

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